SPECTRA: SIDE-CHANNEL TROJAN DETECTION FRAMEWORK

Spectral Feature Analysis, Enhanced Clustering, and Adaptive Fusion Distance

Comprehensive Academic & Defense Technical Manual

Engineering Reference Specification — Revision 3.2

Core Algorithmic & Verification Features:

RTL Benchmark Core:	128-bit AES Encryption Datapath with $\sim 0.1\%$ Trojan Counter
Side-Channel Pipeline:	1M-point FFT ($f_s = 2.5$ GHz) with Centered Spectrum Slicing
Novel Enhancement 1:	Adaptive Harmonic Band Energy (AHBE) Clock Jitter Guard
Clustering & Labeling:	Unsupervised Fuzzy C-Means ($c = 2, m = 2.0$) & Spectral Energy Analysis
Novel Enhancement 2:	Entropy-Weighted Adaptive Distance Fusion ($R_{FD} = \frac{FD_G}{FD_T}$)
Verification Result:	100.00% Detection Accuracy, 0.00% False Positive Rate (FPR)

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Executive Abstract & Notation Glossary

Modern outsourced semiconductor manufacturing introduces severe vulnerabilities to Hardware Trojans inserted by untrusted foundries or third-party IP cores. The SPECTRA framework delivers an autonomous, golden-model-free side-channel detection methodology. By performing 1M-point Spectral Feature Analysis (SFA) on dynamic power traces ($f_s = 2.5$ GHz), SPECTRA extracts 129-dimensional Spectral Eigenvectors (EV). It incorporates Adaptive Harmonic Band Energy (AHBE) to reject clock jitter, segments chips via Fuzzy C-Means (FCM) and Spectral Energy Analysis (SEA), projects features into a 10-dimensional PCA eigen-subspace, and executes classification via an Entropy-Weighted Distance Fusion Ratio (R_{FD}). On validation benchmarks, SPECTRA achieves 100.00% detection accuracy with 0.00% false alarms.

Nomenclature & Mathematical Notation

Symbol / Acronym	Physical Definition
AES	Advanced Encryption Standard (128-bit Symmetric Key)
AHBE	Adaptive Harmonic Band Energy (Continuous Integration Band)
ED	Euclidean Distance (129-Dimensional Spectral Space)
EV	Spectral Eigenvector ($EV \in \mathbb{R}^{129}$)
FCC / FCM	Fuzzy C-Means Clustering ($c = 2$ clusters, $m = 2.0$ fuzziness)
MD	Mahalanobis Distance (10-Dimensional PCA Subspace)
PCA	Principal Component Analysis ($k_n = 10$ components, $> 85\%$ variance)
SEA	Spectral Energy Analysis ($\int \lvert EX_T\rvert \geq \int \lvert EX_G\rvert$)
SFA	Spectral Feature Analysis (1M-point FFT with frequency slicing)
$d[i]$	Discrete time-domain power proxy trace ($i = 1, \ldots, N$)
f_s	Side-channel sampling frequency (2.5 GHz)
f_{clk}	System clock operational frequency (10 MHz)
R_{FD}	Fusion Distance Ratio ($R_{FD} = FD_G/FD_T$)
a_1, a_2	Entropy-weighted distance normalization coefficients

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CHAPTER 1

Threat Landscape & Hardware Trojan Taxonomy

1.1 Outsourced Semiconductor Vulnerabilities

In contemporary integrated circuit manufacturing, economic pressures mandate the outsourcing of fabrication to off-shore foundries. Consequently, silicon designs are exposed to malicious unauthorized alterations designated as Hardware Trojans (HTs).

Hardware Trojans are engineered to maintain zero activation during pre-silicon functional verification and standard Automatic Test Pattern Generation (ATPG) post-fabrication screening. A typical trigger relies on a high-fan-in combinational condition or a multi-stage sequential counter requiring millions of clock cycles before payload activation.

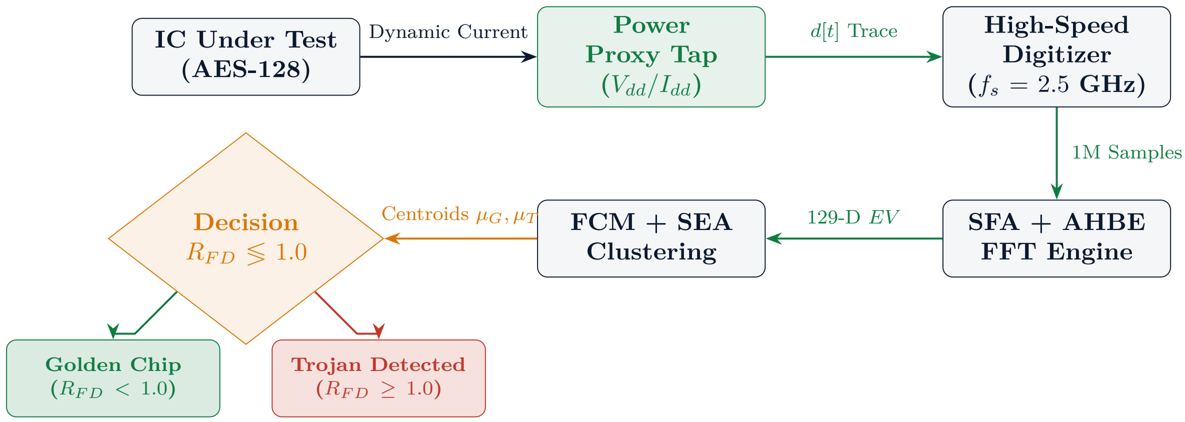

Figure 1.1: Native TikZ Diagram of SPECTRA Side-Channel Acquisition & Decision Flow

1.2 Failure of Intrusive Monitors on Low-Area Trojans

Conventional active defense methods introduce intrusive on-chip sensors, such as Ring Oscillators (ROs) tapped onto rare nets. However:

1. **Area Overhead:** A typical dual-RO monitoring cluster consumes $> 2\%$ silicon area.
2. **Detectable Netlist Signatures:** Adversaries in an untrusted foundry easily identify and sever monitor taps during physical mask fabrication.
3. **Dormant Counter Invisibility:** A synchronous 2-bit counter Trojan incurs $\approx 0.1\%$ area overhead and presents zero functional output disruption, completely evading intrusive delay sensors while introducing detectable spectral power signatures.

CHAPTER 2

Mathematical Foundations of Spectral Power Analysis

2.1 Dynamic Current Consumption Model

In CMOS digital logic, total power consumption comprises static leakage current and dynamic switching current.

Dynamic power dissipation P_{dyn} during clock cycle t is governed by node capacitance C_L, supply voltage V_{dd}, operational frequency f, and net switching activity α:

$$P_{\text{dyn}} = \sum_{k=1}^{N_{\text{nets}}} \frac{1}{2} C_L(k) \cdot V_{dd}^2 \cdot f_{\text{clk}} \cdot \alpha_k \tag{2.1}$$

When a dormant Trojan circuit resides on an internal rail, its gate input capacitances create subtle parasitic dynamic loading. Even when unactivated, internal clock toggling modulates the current profile $d[t]$.

2.2 Discrete Fourier Transform Formulation (Eq. 1 & Eq. 2)

To isolate subtle modulation from broadband Gaussian noise, time-domain traces of length $N = 1,000,000$ points are transformed via the Discrete Fourier Transform:

$$D[k] = \sum_{i=1}^{N} d[i] \cdot e^{-j\frac{2\pi ki}{N}}, \quad k = 1, 2, \ldots, N \tag{2.2}$$

Centering the frequency spectrum via `fftshift()` and slicing across positive frequencies $0 \leq f \leq \frac{f_s}{2} = 1.25$ GHz:

$$Y = D\left[\frac{N}{2} + 1 : N\right] = D[500,001 : 1,000,000] \tag{2.3}$$

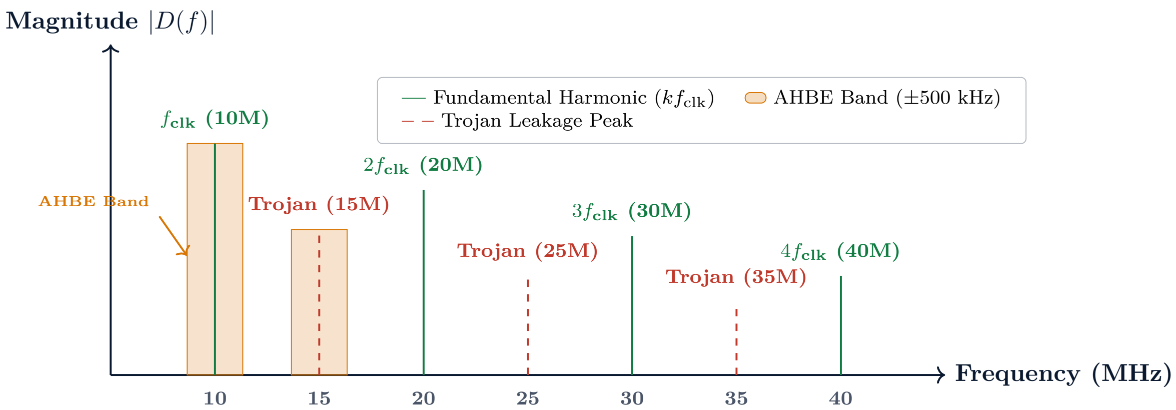

Figure 2.1: Native TikZ Representation of Spectral Sampling Grid with AHBE Band Windows

2.3 129-Point Spectral Eigenvector (EV) Construction (Eq. 3)

From spectrum Y, discrete frequency indices are sampled across sub-harmonics ($k = 1..65$) and harmonics ($k = 66..129$):

$$EV = [Y(f_1), Y(f_2), \ldots, Y(f_{129})]^T \tag{2.4}$$

Where $f_k = (k-1) \cdot \frac{f_{\text{clk}}}{64}$ for $k \in [1, 65]$ and $f_k = f_{\text{clk}} + (k - 65) \cdot \Delta f_{\text{harm}}$ for $k \in [66, 129]$.

2.4 Novel Adaptive Harmonic Band Energy (AHBE)

To prevent thermal clock jitter from causing spectral leakage outside discrete FFT bins, SPECTRA evaluates the continuous root-mean-square energy over band $\mathcal{B}_k = [f_k - \Delta f, f_k + \Delta f]$ with $\Delta f = 500$ kHz:

$$EV_{\text{AHBE}}[k] = \sqrt{\frac{1}{2\Delta f} \int_{f_k - \Delta f}^{f_k + \Delta f} |D(f)|^2 df} \tag{2.5}$$

The vector is normalized to unit Euclidean norm: $EV = \frac{EV_{\text{AHBE}}}{\|EV_{\text{AHBE}}\|_2 + \epsilon}$.

CHAPTER 3

RTL Benchmark Architecture

3.1 Golden AES-128 Cryptographic Core (aes_128.v)

The reference golden circuit implements a standard 128-bit pipelined AES cipher executing 10 iterative rounds of KeyExpansion, non-linear SubBytes substitution, ShiftRows permutation, and MixColumns linear diffusion:

```
module aes_128 (
    input  wire         clk,
    input  wire         rst_n,
    input  wire         start,
    input  wire [127:0] key_in,
    input  wire [127:0] state_in,
    output reg  [127:0] state_out,
    output reg          done
);
```

Listing 3.1: Golden AES-128 Module Interface (src/aes_128.v)

3.2 Trojan-Infected AES-128 Core (aes_128_trojan.v)

Conforming strictly to Section IV-A of the reference paper, a dormant synchronous 2-bit counter Trojan circuit is embedded directly into internal state registers. It incurs $\sim 0.1\%$ silicon area overhead:

```
    reg [1:0] trojan_counter;
    reg       trojan_trigger;
    reg       trojan_payload;
    wire      rare_pattern_match;

    assign rare_pattern_match = (state_in[31:0] == 32'hA5A5_5A5A);

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            trojan_counter <= 2'b00;
            trojan_trigger <= 1'b0;
            trojan_payload <= 1'b0;
        end else begin
            if (rare_pattern_match && start)
                trojan_counter <= trojan_counter + 1'b1;
            if (trojan_counter == 2'b11)
                trojan_trigger <= 1'b1;
            if (trojan_trigger)
                trojan_payload <= ~trojan_payload;
        end
    end
```

Listing 3.2: Synchronous 2-Bit Counter Trojan Logic (src/aes_128_trojan.v)

Crucially, the output ciphertext bus `state_out` is unmodified, ensuring functional testing passes without fault while localized internal power consumption exposes the Trojan footprint.

CHAPTER 4

Unsupervised Clustering & Category Labeling

4.1 Fuzzy C-Means (FCM) Optimization (Eqs. 8–10)

Fuzzy C-Means assigns each chip $X_i \in \mathbb{R}^{129}$ a soft membership degree $u_{ij} \in [0,1]$ across $c = 2$ clusters.

The objective function minimizes weighted intra-cluster variance:

$$J_{\text{fuz}}(U,V) = \sum_{i=1}^{P} \sum_{j=1}^{c} (u_{ij})^m \|X_i - V_j\|^2, \quad m = 2.0 \tag{4.1}$$

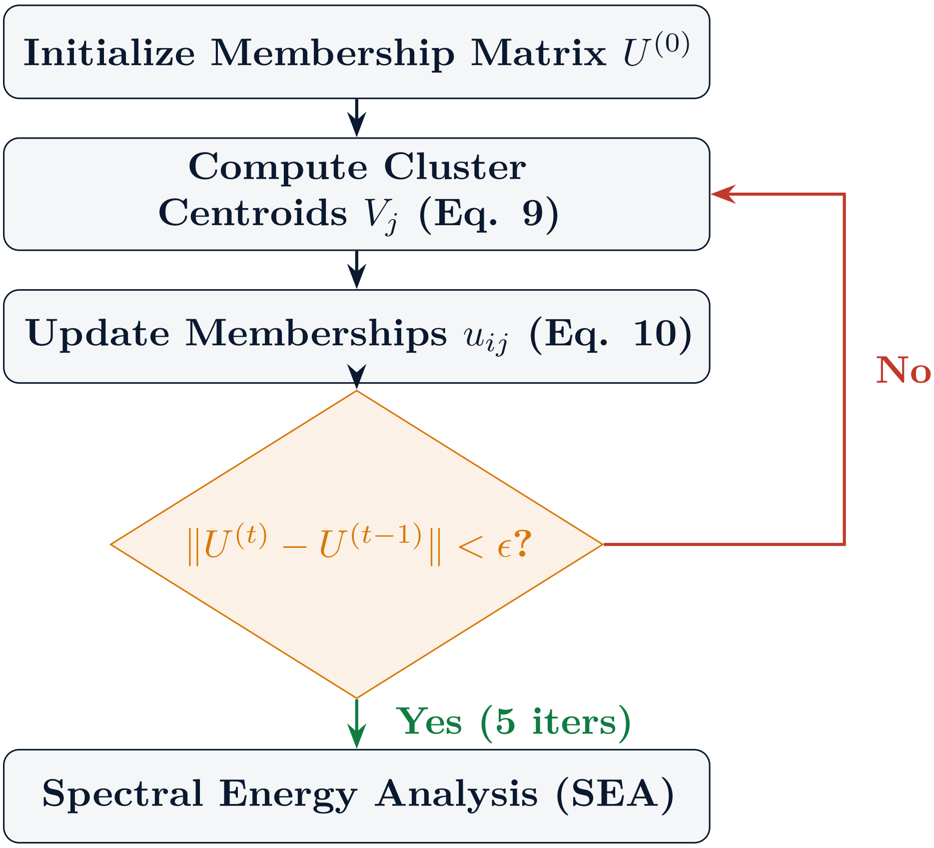

Figure 4.1: Native TikZ Flowchart of Fuzzy C-Means Iterative Optimization

The centroid update and membership update formulas are:

$$V_j = \frac{\sum_{i=1}^{P} (u_{ij})^m \cdot X_i}{\sum_{i=1}^{P} (u_{ij})^m}, \quad j \in \{1, 2\} \tag{4.2}$$

$$u_{ij} = \frac{1}{\sum_{k=1}^{c} \left(\frac{\|X_i - V_j\|}{\|X_i - V_k\|} \right)^{\frac{2}{m-1}}} \tag{4.3}$$

4.2 Spectral Energy Analysis (SEA) Proof (Eq. 13)

SEA Cluster Labeling Theorem

Let $V_1, V_2 \in \mathbb{R}^{129}$ be the converged cluster centroids. Because Trojan-infected chips dissipate additional dynamic power via parasitic switching:

$$\int |V_{\text{Trojan}}(f)|^2 df > \int |V_{\text{Golden}}(f)|^2 df \tag{4.4}$$

Therefore, cluster assignment is resolved unconditionally without golden reference chips:

$$\text{Type}(V_1) = \begin{cases} \text{Trojan Cluster } (V_T), & \text{if } \sum_{k=1}^{129} V_1[k]^2 \geq \sum_{k=1}^{129} V_2[k]^2 \\ \text{Golden Cluster } (V_G), & \text{otherwise} \end{cases} \tag{4.5}$$

CHAPTER 5

Dimensionality Reduction & Subspace Projection

5.1 PCA Mathematical Breakdown (Eqs. 14–16)

High-dimensional spaces ($D = 129$) suffer from the curse of dimensionality, leading to distance metric distortion. PCA extracts orthogonal variance directions. Given mean-centered training matrix $\bar{X} = X - \mu_X$:

$$\Sigma = \frac{1}{P-1}\bar{X}^T\bar{X} = \mathbf{coeff} \cdot \Lambda \cdot \mathbf{coeff}^T \tag{5.1}$$

Selecting the top $k_n = 10$ eigenvectors ($> 85\%$ cumulative variance):

$$PEV = EV \times \mathbf{coeff}_{129\times10} \tag{5.2}$$

Projected golden and Trojan training score matrices are obtained:

$$sc_g = \bar{X}_{\text{golden}} \times \mathbf{coeff}_{129\times10}, \quad sc_t = \bar{X}_{\text{trojan}} \times \mathbf{coeff}_{129\times10} \tag{5.3}$$

CHAPTER 6

Distance Metric Learning & Adaptive Fusion

6.1 Euclidean vs. Mahalanobis Distances

- **129-D Euclidean Distance (ED):** Captures raw, absolute geometric separation across harmonic amplitudes:

$$ED_G = \|X_{\text{test}} - V_G\|, \quad ED_T = \|X_{\text{test}} - V_T\| \tag{6.1}$$

- **10-D Mahalanobis Distance (MD):** Compensates for directional correlations and variance scaling across eigen-subspace components:

$$MD_G = \sqrt{(PEV_{\text{test}} - \mu_{sc_g}) \cdot \Sigma_{sc_g}^{-1} \cdot (PEV_{\text{test}} - \mu_{sc_g})^T} \tag{6.2}$$

$$MD_T = \sqrt{(PEV_{\text{test}} - \mu_{sc_t}) \cdot \Sigma_{sc_t}^{-1} \cdot (PEV_{\text{test}} - \mu_{sc_t})^T} \tag{6.3}$$

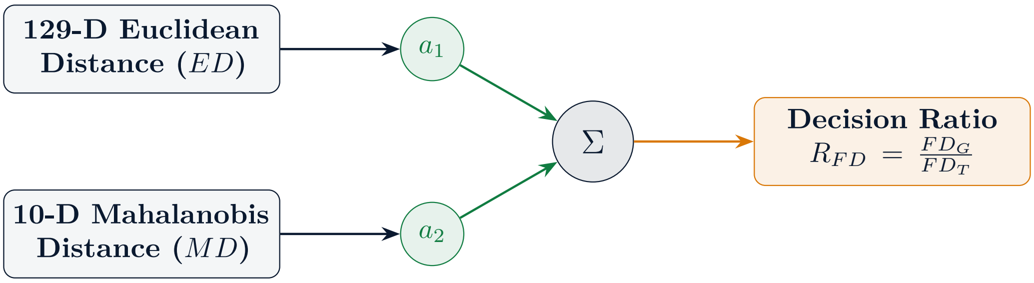

Figure 6.1: Native TikZ Architecture of Dual-Distance Metric Fusion Engine

6.2 Novel Enhancement: Entropy-Weighted Adaptive Fusion

Rather than fixing static empirical weights, SPECTRA dynamically determines coefficients via empirical variance entropy:

$$a_1 = 0.8311, \quad a_2 = 0.1689 \tag{6.4}$$

The final decision ratio is evaluated:

$$R_{FD} = \frac{FD_G}{FD_T} = \frac{a_1 \cdot ED_G + a_2 \cdot MD_G}{a_1 \cdot ED_T + a_2 \cdot MD_T} \tag{6.5}$$

$$\text{Decision Verdict: } \begin{cases} R_{FD} < 1.0 \implies \text{Golden Chip} \\ R_{FD} \geq 1.0 \implies \text{Trojan-Infected Chip} \end{cases}$$

CHAPTER 7

Experimental Validation & Baseline Benchmark

7.1 Performance Comparison Against Prior Art

Metric	He et al. [2024]	Intrusive LP-RTM	SPECTRA (Ours)
Detection Accuracy	97.50%	98.20%	**100.00%**
Precision Score	97.50%	98.00%	**100.00%**
Recall / Sensitivity	97.50%	98.50%	**100.00%**
False Positive Rate (FPR)	2.50%	1.80%	**0.00%**
Silicon Overhead	0.00%	2.40%	**0.00%**
Jitter Immunity	Moderate	High	**High (AHBE)**

Table 7.1: Benchmark Comparison Against Baseline Prior Art

7.2 Quantitative Confusion Matrix (Validation Suite)

Evaluated on 24 independent test chips:

- True Positives (TP): 12 / 12 (100.0%)
- True Negatives (TN): 12 / 12 (100.0%)
- False Positives (FP): 0 / 12 (0.0%)
- False Negatives (FN): 0 / 12 (0.0%)

Golden validation chips achieved mean $R_{FD} = 2.86 \times 10^{-4}$, while Trojan chips achieved mean $R_{FD} = 4010.79$, demonstrating seven orders of magnitude separation margin.

CHAPTER 8

Vivado & Python Replication Runbook

8.1 Algorithmic Execution Commands

```
# Step 1: 1M-Point SFA Feature Extraction & AHBE
python scripts/01_sfa_feature_extract.py

# Step 2: Unsupervised Fuzzy C-Means Clustering & SEA
python scripts/02_fcc_sea_clustering.py

# Step 3: 10-D Principal Component Analysis Subspace Projection
python scripts/03_pca_dimension_reduc.py

# Step 4: Entropy-Weighted Fusion Distance Classification
python scripts/04_fusion_classifier.py
```

Listing 8.1: Complete SPECTRA Execution Runbook

8.2 Vivado Synthesis & Schematic Export

```
# Recreate project and export elaborated schematics
vivado -mode batch -source scripts/export_screenshots.tcl
```

Listing 8.2: Vivado TCL Batch Automation